

GemVAE: A Graph-Enhanced Multi-modal Variational Autoencoder for Spatially Resolved Multi-Omic Data Integration

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Abstract—The emergence of spatially resolved multi-omic technologies enables simultaneous profiling of transcriptomics and proteomics within intact tissue contexts, yet the resulting data pose severe computational challenges: dimensionalities exceeding 30,000 features per modality, inter-modal heterogeneity, and the need to preserve subcellular spatial organization. We propose GemVAE, a Graph-Enhanced Multi-modal Variational Autoencoder that addresses these challenges through three synergistic innovations. First, a dual-stream encoder independently compresses gene expression and protein abundance into modality-specific representations before joint latent projection. Second, an adaptive graph attention mechanism constructs a tissue-topology graph from spatial coordinates and learns attention-weighted cell-cell relationships, enabling the model to encode spatial context directly into the latent embedding. Third, a contrastive learning objective enforces spatial coherence by pulling together representations of spatially adjacent cells. GemVAE reduces heterogeneous multi-omic data to a compact 30-dimensional latent space while preserving both inter-modal dependencies and spatial tissue architecture. Experiments on Stereo-seq SiteSeq mouse thymus and Spatial CITE-seq human breast cancer and spleen datasets demonstrate that GemVAE achieves competitive spatial clustering performance against state-of-the-art methods including STAGATE, GraphST, and SpaGCN, while uniquely supporting joint transcriptomic-proteomic integration at spatial resolution.

Index Terms—spatial transcriptomics, variational autoencoder, graph attention network, multi-omics integration, single-cell genomics, contrastive learning, dimensionality reduction

I. INTRODUCTION

High-throughput sequencing technologies have transformed biological research by enabling genome-scale measurements across individual cells. Spatial transcriptomics extends this capability by preserving the physical location of cells within intact tissues, revealing how gene expression programs are organized in space [9]. Recent platforms such as Stereo-seq [14], 10x Visium, and Spatial CITE-seq [9] further allow simultaneous profiling of transcriptomic and proteomic modalities at subcellular resolution, opening new avenues for understanding tissue heterogeneity, cell-cell communication, and disease microenvironments.

Despite this technological progress, computational analysis of spatially resolved multi-omic data faces three fun-

damental barriers. *First*, the curse of dimensionality [15]: a typical spatial transcriptomics spot profiles 20,000–30,000 genes, while CITE-seq panels add hundreds of protein targets. Standard distance-based algorithms become unreliable in such high-dimensional spaces, and classical dimensionality reduction methods such as Principal Component Analysis (PCA) are limited to linear transformations incapable of capturing the complex nonlinear manifold structure of biological data. *Second*, modality heterogeneity: mRNA counts follow zero-inflated negative binomial distributions, whereas antibody-derived tag (ADT) protein counts exhibit distinct noise profiles and dynamic ranges, demanding separate probabilistic treatment before integration. *Third*, spatial context: cells of the same type may differ in function depending on their tissue location, yet most dimensionality reduction methods treat cells as independent observations, discarding the spatial graph structure entirely.

Existing methods address subsets of these challenges. STAGATE [2] pioneered adaptive graph attention autoencoders for spatial transcriptomics but operates on a single omics modality. scVI [3] and totalVI [4] provide probabilistic frameworks for single-cell and CITE-seq data respectively but lack spatial awareness. GraphST [5] adds self-supervised contrastive learning to graph neural networks for spatial clustering but does not natively handle multi-modal proteomics input. SpaGCN [6] integrates histological images with gene expression via graph convolution but again remains unimodal.

We propose **GemVAE** (Graph-Enhanced Multi-modal Variational Autoencoder), a unified framework that simultaneously handles multi-modal input, spatial graph structure, and high-dimensional data through a variational generative model. Our main contributions are:

- A dual-stream encoder architecture that independently processes transcriptomic and proteomic modalities using BatchNorm-Dropout-Linear blocks with ELU activations before joint latent projection.
- An adaptive bidirectional graph attention mechanism operating on a spatial K-nearest-neighbor graph, learning soft attention weights over cell neighborhoods to encode tissue topology.

- A multi-objective training loss combining MSE or ZINB reconstruction, KL divergence regularization, contrastive spatial coherence, and L2 weight decay, enabling smooth and biologically meaningful latent spaces.
- Comprehensive evaluation on Stereo-seq SiteSeq and Spatial CITE-seq datasets covering mouse thymus, human breast cancer, and spleen tissues.

II. RELATED WORK

A. Variational Autoencoders for Single-Cell Genomics

The Variational Autoencoder (VAE) [8], which learns a latent generative model by maximizing a variational lower bound on the data likelihood, has become a cornerstone of single-cell genomics. Lopez et al. introduced scVI [3], a VAE with negative binomial likelihood that accounts for technical variation across batches in scRNA-seq data. Gayoso et al. extended this to CITE-seq with totalVI [4], using a mixture likelihood to model both RNA and protein counts in a shared latent space. However, neither method incorporates spatial information.

B. Graph Neural Networks for Spatial Transcriptomics

Dong and Zhang proposed STAGATE [2], which combines a graph attention autoencoder with optional cell-type-aware modules to identify spatial domains from 10x Visium data. The adaptive attention mechanism in STAGATE selectively weights contributions from neighboring spots, outperforming fixed-weight graph convolutional approaches. SpaGCN [6] uses graph convolutional networks integrating spatial coordinates and histology images. BayesSpace [7] employs a Markov random field prior for Bayesian subspot-resolution spatial clustering. SEDR [13] learns spatial embeddings through a deep autoencoder with variogram-based spatial regularization.

C. Self-Supervised and Contrastive Spatial Methods

Long et al. proposed GraphST [5], which applies graph self-supervised contrastive learning to pull together representations of spatially adjacent spots while pushing apart non-adjacent ones, achieving state-of-the-art clustering across multiple spatial platforms. DeepST [12] further augments spatial spots with data augmentation strategies before contrastive training. GemVAE extends these ideas to the multi-modal CITE-seq setting, combining modality-specific encoders with spatial graph attention and contrastive objectives.

D. Graph Attention Networks

The Graph Attention Network (GAT) [1] introduced learnable attention coefficients over node neighborhoods, allowing different neighbors to contribute differentially to a node's updated representation. This proved superior to fixed-weight graph convolutional networks (GCNs) on multiple node classification benchmarks. GemVAE adopts a bidirectional dual-path attention mechanism operating on sparse spatial adjacency matrices, extending the original GAT formulation to encoder-decoder autoencoders with variational bottlenecks.

III. METHODOLOGY

A. Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a spatial cell graph where each node $v_i \in \mathcal{V}$ corresponds to a cell or spot with transcriptomic feature vector $\mathbf{x}_i^{(g)} \in \mathbb{R}^{d_g}$ and proteomic feature vector $\mathbf{x}_i^{(p)} \in \mathbb{R}^{d_p}$, and edges $\mathcal{E}$ encode spatial proximity. Here $d_g \approx 20,000$ – $30,000$ and $d_p \approx 100$ – 300 for typical CITE-seq panels. The goal is to learn a compact latent representation $\mathbf{z}_i \in \mathbb{R}^{d_z}$ (with $d_z = 30$) that (1) faithfully reconstructs both modalities, (2) enforces spatial coherence, and (3) supports downstream clustering to delineate tissue spatial domains.

B. Spatial Graph Construction

From the spatial coordinate matrix $\mathbf{S} \in \mathbb{R}^{N \times 2}$ (or $\mathbb{R}^{N \times 3}$ for 3D multi-section data), GemVAE builds a spatial graph using either of two strategies:

K-Nearest Neighbor (KNN): Each cell v_i is connected to its k nearest spatial neighbors using Euclidean distance. The default $k = 6$ balances local and global spatial structure.

Radius: Each cell v_i is connected to all cells within a Euclidean distance threshold r , accommodating irregular spot distributions.

For multi-section 3D data, the method additionally connects cells across adjacent tissue sections using a separate Z-axis radius cutoff, enabling alignment and integration of serial sections. The resulting adjacency is stored as a sparse matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$ with a pruned variant $\hat{\mathbf{A}}$ for the decoder pathway.

C. Dual-Stream Encoder Architecture

GemVAE processes transcriptomic and proteomic inputs through two independent encoder streams. Each stream is composed of stacked BatchNorm-Dropout-Linear (BDL) blocks:

$$\text{BDL}(\mathbf{x}) = \text{Linear}(\text{Dropout}_{0.2}(\text{BN}(\mathbf{x}))) \quad (1)$$

where BN denotes batch normalization. An ELU activation [11] follows each block except the final projection. Let $\text{Enc}_g(\cdot)$ and $\text{Enc}_p(\cdot)$ denote the gene and protein encoder streams respectively, producing intermediate representations $\mathbf{h}_i^{(g)} \in \mathbb{R}^{h_g}$ and $\mathbf{h}_i^{(p)} \in \mathbb{R}^{h_p}$. These are concatenated and jointly projected:

$$\mathbf{h}_i = \text{Linear}([\mathbf{h}_i^{(g)} \parallel \mathbf{h}_i^{(p)}]) \in \mathbb{R}^h \quad (2)$$

Gene encoder dimensions follow $[d_g \rightarrow 512 \rightarrow 256]$ and protein encoder $[d_p \rightarrow 128 \rightarrow 64]$, with the joint projection mapping to a 512-dimensional shared space before the graph attention layers.

D. Bidirectional Graph Attention Mechanism

Given node features $\{\mathbf{h}_i\}$ and the spatial adjacency matrix $\mathbf{A}$, GemVAE computes dual attention pathways f_1 and f_2 using learnable weight vectors $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^{2h}$:

$$f_k(i, j) = \mathbf{a}_k^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j], \quad k \in \{1, 2\} \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{h \times h}$ is a shared linear transformation. The attention logits for each edge $(i, j) \in \mathcal{E}$ are aggregated:

$$s_{ij} = f_1(i, j) + f_2(i, j) \quad (4)$$

and normalized via softmax over each node's local neighborhood:

$$\alpha_{ij} = \frac{\exp(\sigma(s_{ij}))}{\sum_{k \in \mathcal{N}(i)} \exp(\sigma(s_{ik}))} \quad (5)$$

where σ is the sigmoid nonlinearity applied on the sparse adjacency structure. The attended representation aggregates over the spatial neighborhood:

$$\mathbf{h}'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W} \mathbf{h}_j \quad (6)$$

During decoding, the full adjacency $\mathbf{A}$ and a pruned variant $\tilde{\mathbf{A}}$ are blended with an interpolation parameter α_{blend} :

$$\mathbf{A}_{\text{dec}} = \alpha_{\text{blend}} \cdot \mathbf{A} + (1 - \alpha_{\text{blend}}) \cdot \tilde{\mathbf{A}} \quad (7)$$

providing a balance between local and global neighborhood information in the reconstruction pathway.

E. Variational Bottleneck

The graph-attended representation $\mathbf{h}'_i$ is passed through two parallel linear projections to produce the parameters of the variational posterior:

$$\boldsymbol{\mu}_i = \mathbf{W}_\mu \mathbf{h}'_i, \quad \log \sigma_i^2 = \mathbf{W}_\sigma \mathbf{h}'_i \quad (8)$$

Latent codes are sampled via the reparameterization trick [8]:

$$\mathbf{z}_i = \boldsymbol{\mu}_i + \boldsymbol{\epsilon} \odot \boldsymbol{\sigma}_i, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (9)$$

The resulting 30-dimensional latent code $\mathbf{z}_i \in \mathbb{R}^{30}$ captures a disentangled representation of both modalities conditioned on the spatial tissue context.

F. Contrastive Spatial Coherence Loss

To encourage spatially adjacent cells to share similar latent representations, GemVAE employs a contrastive loss with temperature $\tau = 0.5$. For a positive pair (i, j) with $(i, j) \in \mathcal{E}$ and a set of in-batch negatives $\mathcal{N}^-$:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k \in \mathcal{N}^-} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)} \quad (10)$$

where $\text{sim}(\mathbf{u}, \mathbf{v}) = \exp(\mathbf{u}^T \mathbf{v} / \tau)$ measures scaled pairwise similarity.

G. Reconstruction Loss

GemVAE supports two reconstruction objectives. For normalized data, mean squared error:

$$\mathcal{L}_{\text{recon}} = \sqrt{\sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2} \quad (11)$$

For raw count data, a Zero-Inflated Negative Binomial (ZINB) loss models technical dropout prevalent in spatial transcriptomics:

$$p(x | \mu, \theta, \pi) = \pi \cdot \mathbf{1}[x = 0] + (1 - \pi) \cdot \text{NB}(x; \mu, \theta) \quad (12)$$

where μ , θ , and π are the mean, dispersion, and dropout rate predicted by the decoder.

H. Total Training Objective

The model is trained end-to-end by minimizing a weighted sum of four objectives:

$$\mathcal{L} = \lambda_r \mathcal{L}_{\text{recon}} + \lambda_{kl} \mathcal{L}_{\text{KL}} + \lambda_c \mathcal{L}_{\text{contrast}} + \lambda_{wd} \mathcal{L}_{wd} \quad (13)$$

The KL divergence term is:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^{d_z} (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2) \quad (14)$$

and $\mathcal{L}_{wd} = \|\boldsymbol{\Theta}\|_2^2$ is L2 weight decay over all trainable parameters $\boldsymbol{\Theta}$. Default weights: $\lambda_r = 1$, $\lambda_{kl} = 0$ (deterministic mode), $\lambda_c = 10$, $\lambda_{wd} = 1$. The dominance of λ_c prioritizes spatial coherence in the learned embedding. The model is optimized with Adam [10] at learning rate 10^{-4} with gradient clipping at norm 5, for 500 epochs.

IV. EXPERIMENTS

A. Datasets

We evaluate GemVAE on four spatially resolved multi-omic datasets spanning two technologies and multiple tissue types:

Stereo-seq SiteSeq Mouse Thymus (SSC-MT): A high-resolution Stereo-seq dataset [14] profiling the mouse thymus at subcellular resolution, capturing transcriptomic profiles alongside spatial barcoding. This dataset tests the model's ability to resolve fine-grained immune compartments within a complex lymphoid organ.

Spatial CITE-seq Breast Cancer (BC): Human breast tumor tissue [9] simultaneously profiled for whole transcriptome and surface protein markers via antibody-derived tags (ADTs). The tumor microenvironment, including cancer cells, stromal cells, and immune infiltrates, provides a challenging multi-domain clustering target.

Spatial CITE-seq Spleen (Replicates 1 and 2): Two serial sections of human spleen providing paired RNA and protein measurements. Replicate structure allows assessment of cross-replicate consistency in predicted spatial domains.

All datasets were preprocessed with library-size normalization and log1p transformation, followed by highly variable gene selection (top 3,000 genes). Protein ADT counts were independently normalized. Spatial graphs were constructed using the KNN model ($k = 6$) from tissue coordinates.

B. Baseline Methods

GemVAE is benchmarked against six methods spanning different methodological families:

- **PCA + mclust**: Linear dimensionality reduction followed by model-based Gaussian clustering [16]; serves as a simple non-spatial baseline.
- **totalVI** [4]: Multi-modal CITE-seq VAE operating without spatial graph; isolates the effect of spatial modeling.
- **SpaGCN** [6]: GCN-based spatial integration using expression and histological image features.
- **BayesSpace** [7]: Markov random field Bayesian method for spatially-aware clustering.
- **STAGATE** [2]: Adaptive graph attention autoencoder; the closest single-modality predecessor to GemVAE.
- **GraphST** [5]: Graph self-supervised contrastive learning; current state-of-the-art for single-modality spatial clustering.

C. Implementation Details

GemVAE is implemented in TensorFlow. Gene encoder hidden dimensions: [512, 256]; protein encoder hidden dimensions: [128, 64]; joint latent dimension $d_z = 30$. Dropout rate: 0.2 in each BDL block. Adam optimizer [10], learning rate 10^{-4} , gradient clip norm 5, 500 training epochs. Spatial graph: KNN with $k = 6$. Contrastive temperature $\tau = 0.5$. Clustering performed using mclust [16] on the 30-dimensional latent embeddings, with Leiden and Louvain as optional alternatives. All experiments conducted with random seed 2020 for reproducibility.

D. Evaluation Metrics

Clustering quality is quantified by two metrics:

Adjusted Rand Index (ARI): Agreement between predicted and ground-truth cluster labels, corrected for chance. Range $[-1, 1]$; higher is better.

Normalized Mutual Information (NMI): Shared information between prediction and annotation, normalized by marginal entropies. Range $[0, 1]$; higher is better.

E. Quantitative Results

Table I reports ARI and NMI for all methods across the four datasets. GemVAE achieves the highest scores on all benchmarks. The advantage is most pronounced on the Spatial CITE-seq datasets (BC, Spl-R1, Spl-R2), where GemVAE’s protein modality gives it structural information unavailable to RNA-only baselines. On SSC-MT, which contains only transcriptomic data, GemVAE’s spatial graph attention and contrastive objective still outperform STAGATE and GraphST by incorporating both the spatial graph and the variational regularization simultaneously.

TABLE I
CLUSTERING PERFORMANCE (ARI / NMI) ACROSS DATASETS

Method	SSC-MT	BC	Spl-R1	Spl-R2
PCA	0.38 / 0.52	0.32 / 0.48	0.35 / 0.50	0.33 / 0.49
SpaGCN	0.44 / 0.58	0.38 / 0.53	0.42 / 0.56	0.40 / 0.55
BayesSpace	0.47 / 0.61	0.41 / 0.57	0.44 / 0.59	0.43 / 0.58
totalVI	0.41 / 0.55	0.45 / 0.60	0.43 / 0.58	0.42 / 0.57
STAGATE	0.56 / 0.67	0.49 / 0.63	0.53 / 0.64	0.51 / 0.63
GraphST	0.59 / 0.70	0.52 / 0.65	0.57 / 0.68	0.55 / 0.67
GemVAE	0.62 / 0.73	0.67 / 0.76	0.64 / 0.74	0.63 / 0.73

BC = Breast Cancer; Spl = Spleen replicates. Best scores **bolded**. GemVAE gains largest margin on CITE-seq datasets where single-modality baselines cannot leverage protein information.

F. Ablation Study

Table II quantifies the contribution of each architectural component on the Breast Cancer dataset by ablating one component at a time:

TABLE II
ABLATION STUDY ON BREAST CANCER DATASET

Variant	ARI	NMI
GemVAE-noGraph	0.48	0.61
GemVAE-noProtein	0.53	0.65
GemVAE-noContrast	0.55	0.67
GemVAE (Full)	0.67	0.76

GemVAE-noGraph: spatial adjacency removed; GemVAE-noProtein: gene encoder only; GemVAE-noContrast: $\lambda_c = 0$.

Removing the spatial graph produces the largest performance drop ($\Delta\text{ARI} = -0.19$), confirming that tissue-topology-aware attention is the primary driver. Removing the protein modality costs $\Delta\text{ARI} = -0.14$, and removing the contrastive objective costs $\Delta\text{ARI} = -0.12$. All three components are necessary for peak performance.

G. Visualization

UMAP projections of GemVAE’s 30-dimensional latent embeddings yield well-separated clusters corresponding to annotated tissue compartments. When overlaid on tissue section scatter plots, predicted spatial domains align strongly with known anatomical boundaries, including cortex versus medulla in the thymus and tumor core versus stromal regions in breast cancer.

V. DISCUSSION

GemVAE demonstrates that integrating spatial graph attention with multi-modal variational inference yields substantially richer cell representations than either approach alone. The dual-stream encoder effectively handles the distributional mismatch between RNA counts and ADT protein measurements by compressing each modality independently before joint integration. The bidirectional graph attention mechanism captures soft, continuously-varying spatial relationships without requiring predefined domain boundaries.

A notable design choice is the contrastive weight $\lambda_c = 10$, which strongly prioritizes spatial coherence in the latent space.

This is beneficial for clustering but may reduce reconstruction fidelity for imputation-focused applications; users can tune λ_c downward to shift this balance. Similarly, setting $\lambda_{kl} = 0$ (deterministic mode) was found to produce better-clustered embeddings than the full stochastic VAE, consistent with observations in STAGATE and GraphST.

The 3D extension via Z-axis graph connections opens possibilities for integrating serial tissue sections, enabling volumetric spatial domain identification. Future directions include: (1) interpretable attention weight visualization for gene-protein co-regulation analysis; (2) incorporation of chromatin accessibility (scATAC-seq) as a third modality; (3) PyTorch reimplementation for GPU acceleration on large-scale Stereo-seq data with millions of bins.

VI. CONCLUSION

We presented GemVAE, a Graph-Enhanced Multi-modal Variational Autoencoder for spatially resolved multi-omic data integration. By unifying dual-stream modality-specific encoders, bidirectional spatial graph attention, and a contrastive coherence objective within a single variational framework, GemVAE addresses the joint challenges of extreme dimensionality, modality heterogeneity, and spatial context preservation. Quantitative evaluation across four datasets and an ablation study confirm that each architectural component contributes measurably to downstream spatial clustering quality. GemVAE represents a step toward fully integrated, spatially-aware multi-omic analysis tools that can reveal the functional organization of complex tissues at subcellular resolution.

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